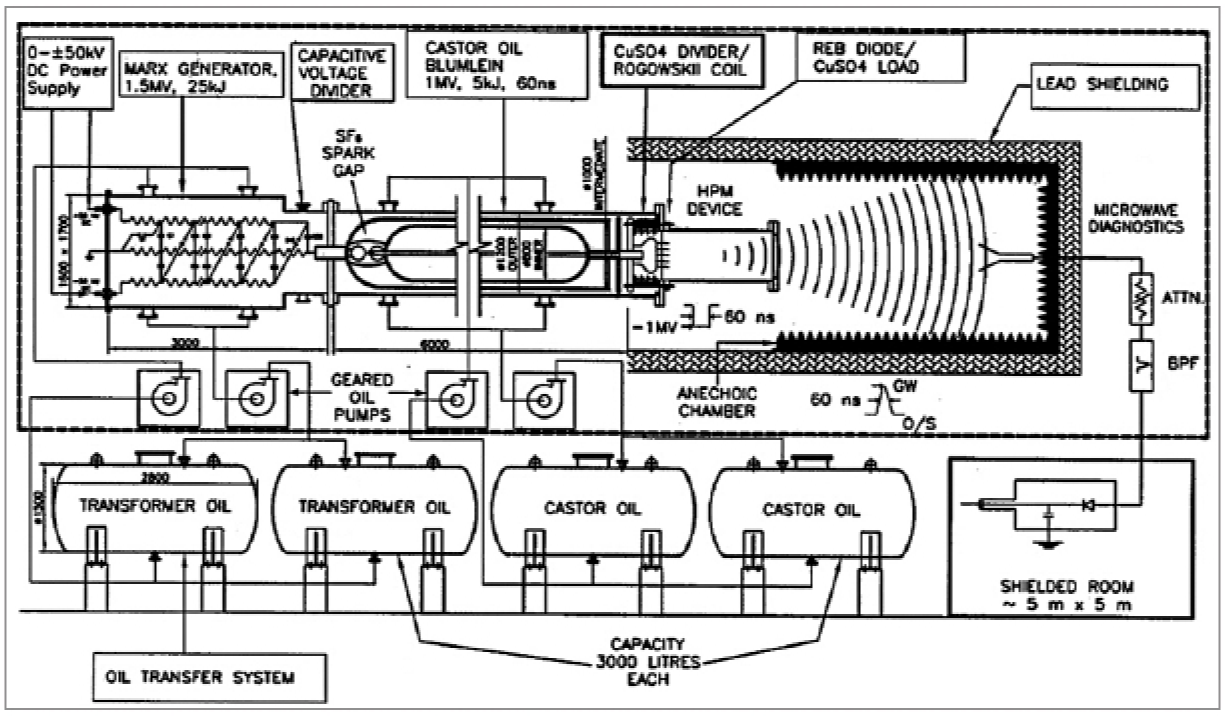

Fig. 8: Schematic diagram of a pulsed linear accelarator weapon

Fig. 9: KALI 5000 system

into the armature and then back along the other rail. It is based on principles similar to those of homopolar motor. However, the problem is that ships, like an aircraft carrier or destroyer, which can generate 25MW of power, are only suitable for its installation due to the high power requirement of the rail gun. The device uses super-capacitors to store the enormous power and deliver it as a pulse in a short period of time. Fig. 7 shows the arrangement of a railgun.

Heavy mobility truck. Heavy Expanded Mobility Tactical Truck (HEMTT) is an eight-wheel-drive, 9,100kg tactical truck with a 470hp diesel engine used by the US army and many other countries for transporting combat vehicles and weapons. This diesel-electric hybrid truck does not use batteries. Instead, it uses supercapacitors that accept more of the power created by regenerative braking and deliver the power to the wheels during acceleration. This increases efficiency by up to 20 per cent. This hybrid technology delivers up to 120kW of power to run an airfield, weapons, radar systems, hospitals, or a disaster relief command centre.

Pulsed linear accelerator weapon. It is a futuristic directed-energy weapon designed to work in such a way that if any enemy missile is launched towards a country, it will quickly emit powerful pulses of the relativistic electron beam, which is converted by other components of the machine down the line into flash x-ray and high power microwave that destroy the target. Unlike laser beams, it does not bore a hole in the target but thoroughly destroys the onboard electronic system.

As shown in Fig. 8, the circuit generates a high-voltage pulse by a Marx generator by charging several capacitors (likely supercapacitors) in parallel by a DC power supply through the resistors and discharging them in series. The output is available as a brief pulse. USA, Russia, China, and India have this powerful weapon. India has developed this machine and christened it as KALI (Kilo Ampere Linear Injector) (refer Fig. 9). KALI 5000 system can produce electron pulses of about 100ns with energy of about 1MeV, current 40kA, a power of 40GW, and microwave radiation at 3-5GHz. EFY